



HOST INTERFACE + CONTROL EXPANDER

HOST INTERFACE – provisional (connector family NOT frozen; see hardware/pcb/ICD_PERCEPTION_HEAD.md)

J1 = single locking host cable. JST GH 1.25 mm 14–pin chosen for Rev A bench: locking, 1 A/contact, cheap prebuilt cables.

J1 pin order is a BENCH/MECH decision: re–order freely until the host–carrier connector is frozen.

Host–side mapping (Rev A bench harness, UNO Q, NOT final pins):

I2C_A = UNO Q Qwiic I2C4 PD12/PD13 (2.2k pull–ups on UNO Q) I2C_B = JDIGITAL D21/D20 I2C2 PB10/PB11 (TBD_HOST_I2C_B)

TOF_INT_N, TOF_SYNC, IMU_INT1 –> TBD_INT on JDIGITAL 3.3 V pins ILLUM_SET –> TBD_PWM or A0/A1 DAC (STM32 DAC1)

+5V_HOST –> JANALOG 5V (BENCH_VERIFY: present when UNO Q is powered from DC_IN?) +3V3_HOST –> Qwiic/JANALOG 3V3

J2 Qwiic = bench/daisy–chain path for bus A only. Never connect J1 and J2 to two different hosts (3V3 rails would tie).

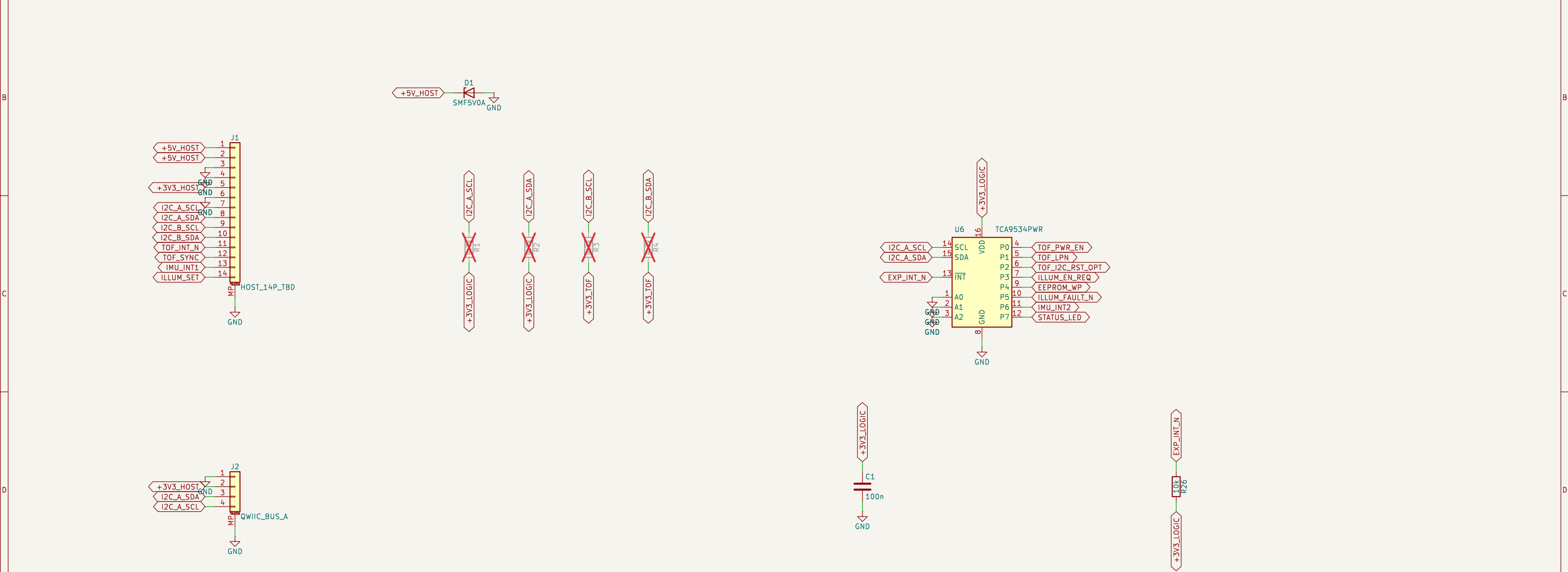
R1/R2 bus–A pull–ups DNP_OPTION: host already has 2.2k. R3/R4 bus–B pull–ups DNP_OPTION: Pololu shifter has 10k to VIN.

U6 TCA9534 @0x20 (A2..A0 = GND). NO internal pull–ups (TCA9534 != TCA9554): external resistors define safe power–up states.

FIRMWARE RULE: write Output reg (0x01) = 0x00 BEFORE setting Config reg (0x03); POR default output reg is 0xFF.

FIRMWARE RULE: P1 (TOF_LPN) is drive–low–only: leave it an INPUT (carrier pulls LP high); to assert, output 0.

FIRMWARE RULE: host drives TOF_SYNC open–drain (STM32 GPIO_MODE_OUTPUT_OD), never push–pull (ICD sec 5a).

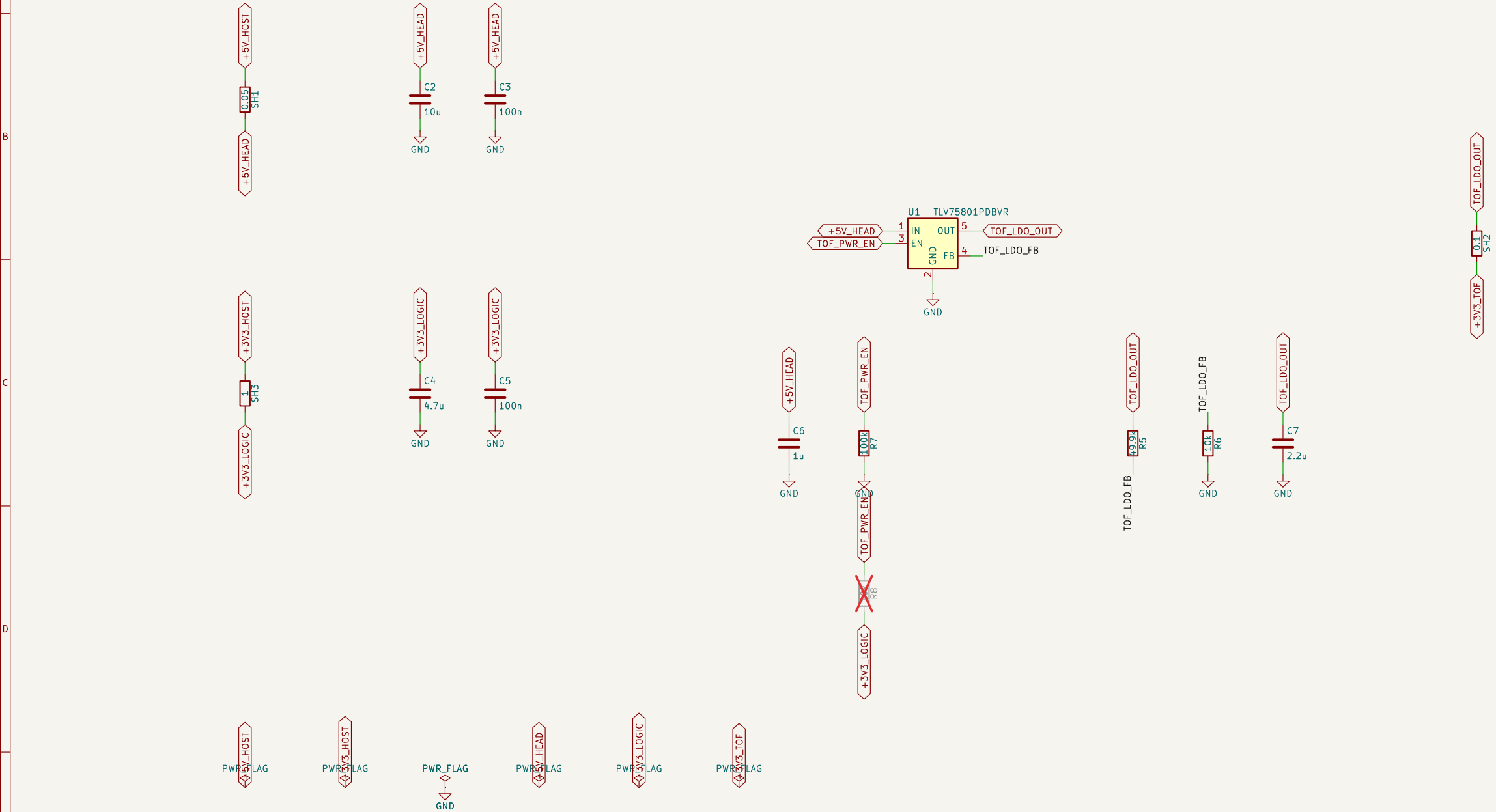


PRE–LAYOUT: not for fabrication. Provisional nets/values are labelled.
Host interface + control expander
Platypus One
Sheet: /HOST_INTERFACE/
File: host_interface.kicad_sch

Title: Perception Head Rev A – HOST_INTERFACE		
Size: A3	Date: 2026–10–05	Rev: A0–prelim
KiCad E.D.A. 9.0.9		Id: 2/8

POWER ENTRY, MEASUREMENT SHUNTS, SWITCHABLE TOF RAIL

POWER – rails come from the host; the head makes only the switchable ToF rail.
+5V_HOST –(SH1 0.05R)–> +5V_HEAD : feeds ToF LDO and illumination switch. Head 5 V current = V(SH1)/0.05.
+3V3_HOST –(SH3 1R)–> +3V3_LOGIC : IMU, EEPROM, expander, op–amp. Logic current = V(SH3)/1 (expect low mA).
U1 TLV75801P –> TOF_LDO_OUT –(SH2 0.1R)–> +3V3_TOF : Pololu VIN. Off by default (R7 pulls EN low).
 $V_{out} = 0.55\text{ V} \times (1 + R5/R6) = 0.55 \times (1 + 49.9k/10k) = 3.29\text{ V}$ nominal (3.23–3.36 V: 1% Vref + 0.5% R5/R6).
Why 3.30 V: Pololu host–side I/O follows VIN, so VIN stays in the host’s 3.3 V logic domain (same condition as the Lab rig).
Every bus–B/INT/SYNC crossing then stays within STM32 VDD+0.3 V and NX50108 VIH = VCC(B)–0.4 V (ICD sec 5a).
BENCH_VERIFY AVDD >= 3.13 V during 8x8 ranging (T2). 3.43 V (R5 = 52.3k) is a CONDITIONAL fallback only (ICD sec 5a).
Why a switchable rail: VL53LBCX reset = remove all supplies for 10 ms (ST UM3109 sec 4.2). TLV758P has active discharge.
R8 DNP_OPTION ‘TOF_FORCE_ON’: populate to power ToF without firmware (bench only).
PWR_FLAGS mark rails driven through shunts/connectors so ERC can check power–pin drive.



PRE–LAYOUT: not for fabrication. Provisional nets/values are labelled.
Power entry, measurement shunts, switchable ToF rail

Platypus One

Sheet: /POWER/
File: power.kicad_sch

Title: Perception Head Rev A – POWER

Size: A3
KiCad E.D.A. 9.0.9

Date: 2026–10–05

Rev: A0–prelim

Id: 3/8

VL53L8CX TOF – REV A HOSTS THE POLOLU #3419 CARRIER

TOF – DECISION B: host the proven Pololu #3419 carrier, not the bare VL53L8CX (rationale: hardware/pcb/RESEARCH_LEDGER.md).

Carrier regulates AVDD 3.3 V + CORE/IOVDD 1.8 V and level-shifts all I/O to VIN (NXS0108). VIN = +3V3_TOF (3.29 V).

Dedicated bus I2C_B: Pololu warns its shifter is load-sensitive (keep wires < 8 cm, few devices) and the

ToF rail is power-cycled for reset – isolating it keeps IMU/EEPROM on bus A alive during ToF recovery.

Address 0x29 (7-bit; 0x52 8-bit). Firmware (~84 KB) is uploaded by the ULD at every power-up.

SPI/I2C pin strapped LOW by R9 (I2C mode). R10 DNP_OPTION lets expander P2 toggle it (I2C-interface reset only).

R28 100k keeps expander P2 defined while R10 is not fitted (if R10 is fitted, firmware must drive P2 low).

MISO and CS unused in I2C mode (carrier pulls CS high). SPI is a Rev B option, not wired.

Bus B crosses the host cable: C18/C19 DNP_OPTION (22 pF) per Pololu note on shifter oscillation; scope it (BENCH_VERIFY).

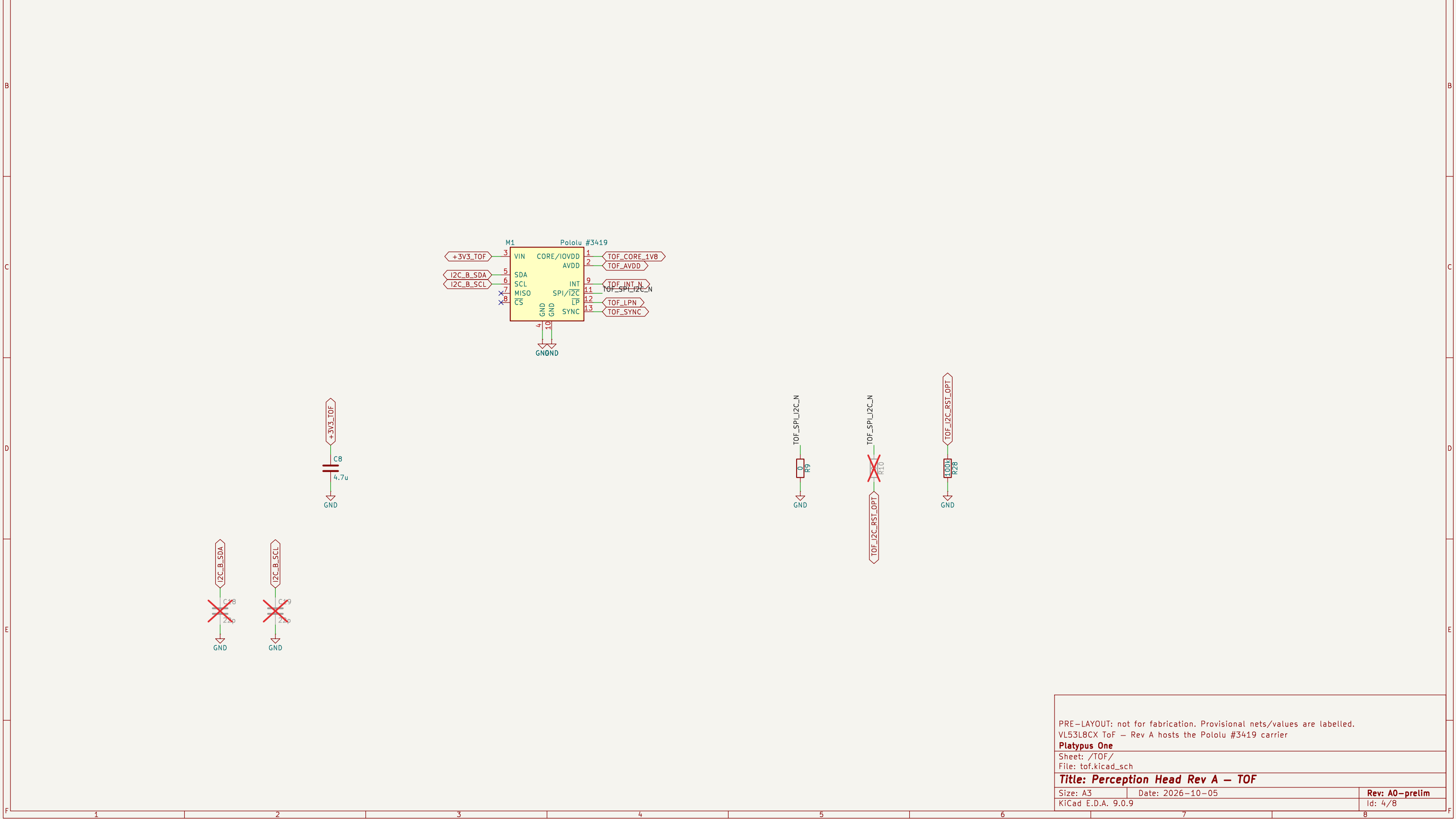
TOF_SYNC: host-driven ranging trigger for camera/ToF timing experiments. HOST DRIVES IT OPEN-DRAIN (high level then comes from the shifter's own pull-up to VIN, so NX50108 VIH = VCC(B)-0.4 V holds by construction). Hi-Z while ToF off.

TOF_LPN (expander P1): drive-low-only (output reg 0, toggle direction) – never push 3.3 V high into the shifter.

AVDD and CORE/IOVDD carrier outputs go only to test points (BENCH_VERIFY AVDD >= 3.13 V during 8x8 ranging).

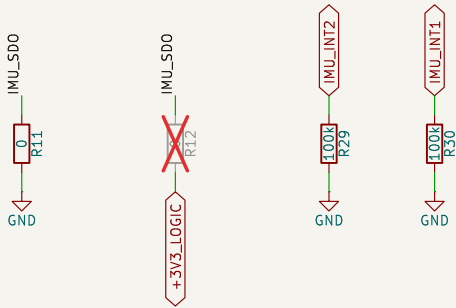
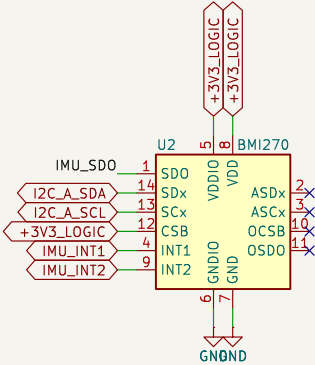
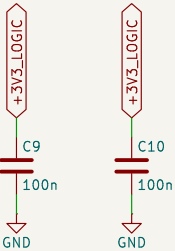
MECHANICAL: carrier is soldered on headers + 2x M2 to the head PCB; its pose is part of the camera $\leftrightarrow$ ToF extrinsic.

BENCH_VERIFY: 1x4 header pin order (GND, SPI/I2C, LP, SYNC) by continuity before layout freeze.



BMI270 IMU

IMU – Bosch BMI270 on bus A (I2C). VDD = VDDIO = +3V3_LOGIC (both inside 1.71/1.2–3.6 V).
CSB tied to VDDIO = I2C mode (Bosch DS sec 6.3). SDO -> GND = 0x68 (R11). R12 DNP_OPTION -> 0x69.
ASDx/ASCx/OCsB/OSDO not connected (DNC), as in Bosch I2C connection diagram 7.2.3 and allowed by DS Table 22.
INT1 -> host (time-critical, direct). INT2 -> expander P6 (non-critical status).
R30/R29 100k pull-downs define INT1/INT2 before firmware enables the outputs (BMI270 INT outputs start disabled);
firmware must configure INT1/INT2 push-pull, active-high.
100 nF on VDD and on VDDIO per Bosch connection diagram 7.2.3; place within 1–2 mm of pins.
PLACEMENT (layout): rigid area near camera/ToF datum, away from illumination BJTs/LDO heat (>=10 mm),
away from mounting screws/board flex zones; no vias in pad, no silkscreen under part; land per Bosch sec 8.3.
Firmware: 8 kB config upload after every POR (Bosch DS sec 4.4); allow >=2 ms supply-to-interface.

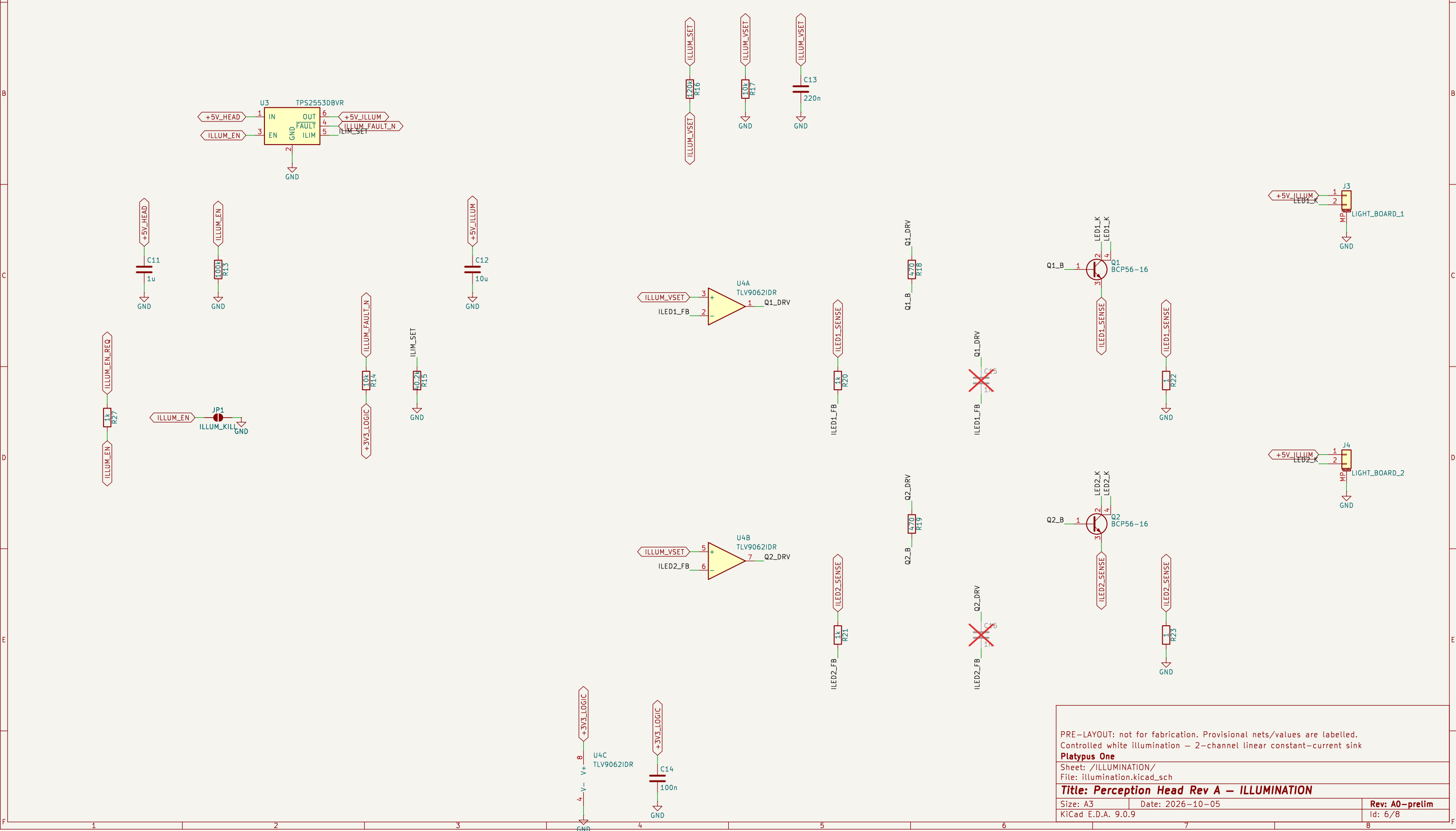


PRE-LAYOUT: not for fabrication. Provisional nets/values are labelled.
BMI270 IMU
Platypus One
Sheet: /IMU/
File: imu.kicad_sch

Title: Perception Head Rev A – IMU		
Size: A3	Date: 2026-10-05	Rev: A0-prelim
KiCad E.D.A. 9.0.9		
Id: 5/8		

CONTROLLED WHITE ILLUMINATION – 2-CHANNEL LINEAR CONSTANT-CURRENT SINK

ILLUMINATION – LED current NOT frozen. Topology frozen–candidate: switched 5 V + linear constant–current sinks.
Why linear CC (not resistor, not buck): brightness independent of 5 V droop and LED Vf; no switching noise next to camera/ToF; analog set–point means NO PWM banding on the IMX219 rolling shutter (PWM–dim only as last resort).
 $I_{LED(per\ ch)} = V(ILLUM_VSET) / 1\ ohm$. $ILLUM_VSET = ILLUM_SET \times 10k / (120k + 10k)$: 3.3 V -> 254 mA full scale.
 $ILLUM_SET$ source: STM32 DAC (A0/A1) preferred, or TBD_PWM >= 20 kHz filtered by R17/C13 (tau ~2 ms).
Off paths (defence in depth): $ILLUM_SET$ low/floating -> R17 pulls VSET to 0; $ILLUM_EN$ low -> U3 off (R13);
JP1 closed -> $ILLUM_EN$ hard–grounded (illumination kill); R27 1k keeps the expander from fighting JP1.
FIRMWARE RULE: set $ILLUM_SET = 0$ before toggling $ILLUM_EN$, then ramp (op–amps saturate while U3 is off).
U3 TPS2553 limit: $R15 = 40.2k$ -> $I_{OS} = 592 / 647 / 709\ mA\ min/nom/max$ (SLVS841F eq.1) > 2 x 254 mA. Protects host 5 V.
Q1/Q2 dissipation at 254 mA: $(5.0 - V_f - 3.0 - 0.25) \times 0.25 = \sim 0.45\ W$ each continuous -> SOT–223 + copper; BENCH thermal.
LEDs live on small off–board light boards (J3/J4) so position/angle/diffuser can iterate without respinning the head.
LED part NOT selected (high–CRI mid–power candidates in BOM). Keep LEDs outside camera FOV cone; symmetric about axis.
C15/C16 DNP_OPTION loop compensation if the op–amp/BJT loop rings (BENCH_VERIFY with scope on SENSE nodes).



PRE-LAYOUT: not for fabrication. Provisional nets/values are labelled.
Controlled white illumination – 2-channel linear constant-current sink

Platypus One

Sheet: /ILLUMINATION/
File: illumination.kicad_sch

Title: Perception Head Rev A – ILLUMINATION

Size: A3	Date: 2026-10-05	Rev: A0-prelim
KiCad E.D.A. 9.0.9		Id: 6/8

BOARD IDENTITY + CALIBRATION STORAGE

ID / CALIBRATION – smallest robust option: one 32–Kbit EEPROM with a factory 128–bit unique serial number.
AT24CS32 on bus A: array at 0x50, read–only serial number in the ‘separate 0x58 ‘security’ address space
(read: dummy write word address 0x0800 to 0x58, then read 16 bytes – Atmel–8869C sec 10).
WP pulled HIGH by R24 = write–protected by default. Host enables writes via expander P4 (EEPROM_WP low).
Contents: compact versioned record (schema id, board rev, assembly variant, camera id, ToF carrier id,
extrinsic camera<–>ToF, intrinsics reference/hash, calibration rev/date, test state) + CRC,
plus the 776–byte VL53LBCX crosstalk buffer (ST AN5939: host stores it, loads it at every start–up).
Full calibration files stay on the host, keyed by the serial number; EEPROM holds identity + compact pose.

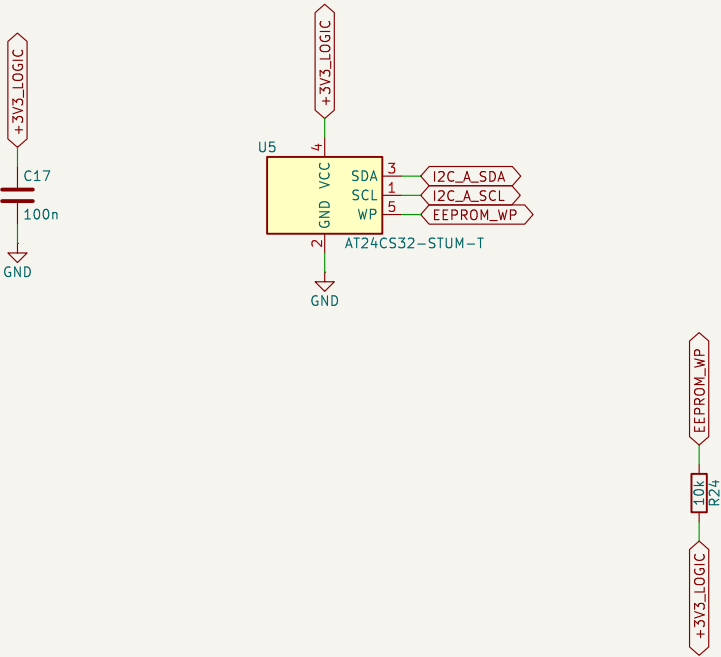
B

C

D

E

F



PRE–LAYOUT: not for fabrication. Provisional nets/values are labelled.
Board identity + calibration storage
Platypus One
Sheet: /ID_CALIBRATION/
File: id_calibration.kicad_sch

Title: Perception Head Rev A – ID_CALIBRATION		
Size: A3	Date: 2026–10–05	Rev: A0–prelim
KiCad E.D.A. 9.0.9		Id: 7/8

TEST POINTS, STATUS LED, MECHANICAL DATUMS

DEBUG / TEST – every rail, every bus and every control/interrupt line has a test point.
Shunt Kelvin pairs: SH1 = TP1/TP2, SH3 = TP4/TP5, SH2 = TP6/TP7; LED current = TP27/TP28 to GND (1 ohm).
GND loops TP10–TP12 spread across the board for scope ground clips.
Layout rules (later): TPs on one side, labelled, >= 2.54 mm apart, reachable with the head assembled to its bracket.
MECHANICAL: H1/H2/H3 implement a 3–2–1 style locate–then–clamp scheme to the sensor datum bracket;
H4–H7 hold the camera module so the head PCB is the rigid camera<–>ToF datum. ALL positions come from Fusion.
Board marking (layout): ‘PLATYPUS ONE PERCEPTION HEAD REV A’, date code, serial label area, pin–1 marks on J1–J4.

